

# Machine Learning and Linear Models

## Least-Squares Lines

design matrix

parameter vector

observation vector

$$X\beta = y$$

This is  
 $Ax = b$

$$X\beta = y, \quad \text{where } X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \quad \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}, \quad y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

The least-squares line  
Also called a line of regression of  $y$  on  $x$

In Figure 1, if the data points were on the line, then.....

| Predicted y-value       |   | Observed y-value |
|-------------------------|---|------------------|
| $\beta_0 + \beta_1 x_1$ | = | $y_1$            |
| $\beta_0 + \beta_1 x_2$ | = | $y_2$            |
| $\vdots$                |   | $\vdots$         |
| $\beta_0 + \beta_1 x_n$ | = | $y_n$            |

regression coefficients:  
 $\beta_0$  and  $\beta_1$

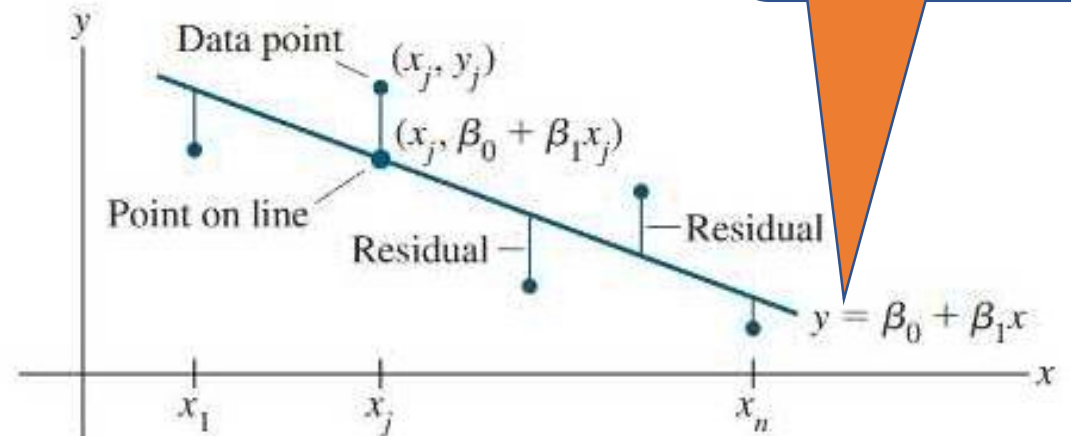


FIGURE 1 Fitting a line to experimental data.

**EXAMPLE 1** Find the equation  $y = \beta_0 + \beta_1 x$  of the least-squares line that best fits the data points  $(2, 1)$ ,  $(5, 2)$ ,  $(7, 3)$ , and  $(8, 3)$ .

**SOLUTION** Use the  $x$ -coordinates of the data to build the design matrix  $X$  in (1) and the  $y$ -coordinates to build the observation vector  $\mathbf{y}$ :

$$X = \begin{bmatrix} 1 & 2 \\ 1 & 5 \\ 1 & 7 \\ 1 & 8 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 3 \end{bmatrix}$$

For the least-squares solution of  $X\boldsymbol{\beta} = \mathbf{y}$ , obtain the normal equations (with the new notation):

$$X^T X \boldsymbol{\beta} = X^T \mathbf{y}$$

That is, compute

$$X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 5 & 7 & 8 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 1 & 5 \\ 1 & 7 \\ 1 & 8 \end{bmatrix} = \begin{bmatrix} 4 & 22 \\ 22 & 142 \end{bmatrix}$$

$$X^T \mathbf{y} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 2 & 5 & 7 & 8 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 3 \end{bmatrix} = \begin{bmatrix} 9 \\ 57 \end{bmatrix}$$

The normal equations are

$$\begin{bmatrix} 4 & 22 \\ 22 & 142 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} 9 \\ 57 \end{bmatrix}$$

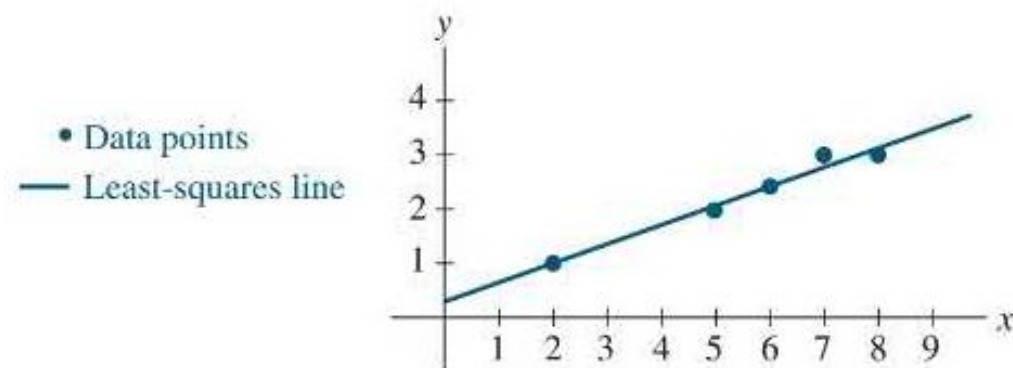
Hence

$$\begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} 4 & 22 \\ 22 & 142 \end{bmatrix}^{-1} \begin{bmatrix} 9 \\ 57 \end{bmatrix} = \frac{1}{84} \begin{bmatrix} 142 & -22 \\ -22 & 4 \end{bmatrix} \begin{bmatrix} 9 \\ 57 \end{bmatrix} = \frac{1}{84} \begin{bmatrix} 24 \\ 30 \end{bmatrix} = \begin{bmatrix} 2/7 \\ 5/14 \end{bmatrix}$$

Thus the least-squares line has the equation

$$y = \frac{2}{7} + \frac{5}{14}x$$

See Figure 2.



**FIGURE 2** The least-squares line  $y = \frac{2}{7} + \frac{5}{14}x$ .

**EXAMPLE 2** If a machine learns the data from Example 1 by creating a least-squares line, what outcome will it predict for the inputs 4 and 6?

**SOLUTION** The machine would perform the same calculations as in Example 1 to arrive at the least-squares line

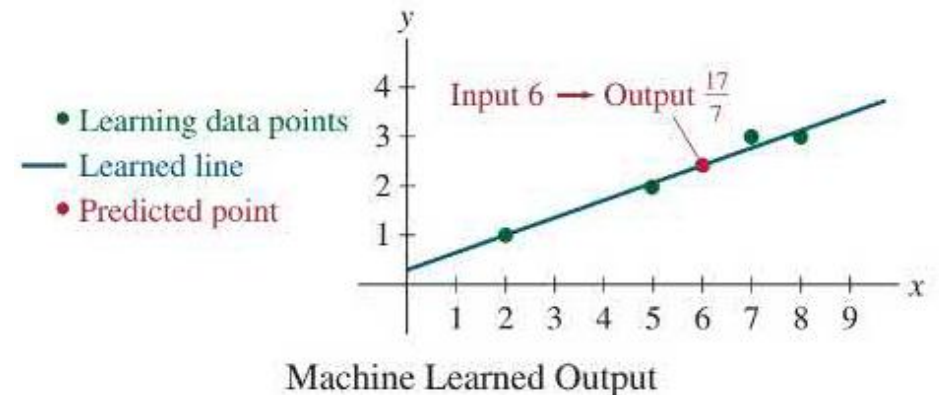
$$y = \frac{2}{7} + \frac{5}{14}x$$

as a reasonable pattern to use to predict the outcomes.

For the value  $x = 4$ , the machine will predict an output of  $y = \frac{2}{7} + \frac{5}{14}(4) = \frac{12}{7}$ .

For the value  $x = 6$ , the machine will predict an output of  $y = \frac{2}{7} + \frac{5}{14}(6) = \frac{17}{7}$ .

See Figure 3.

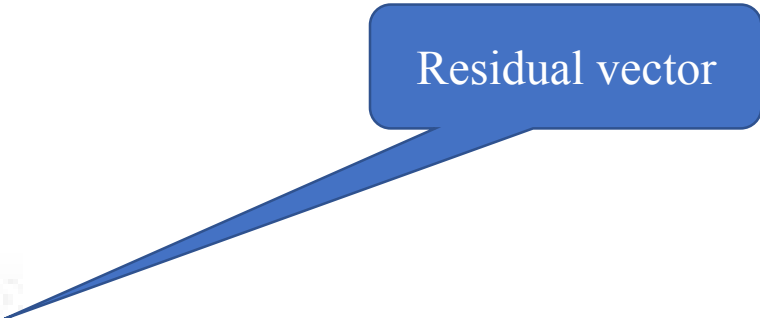


**FIGURE 3** Machine-learned output.

# The General Linear Model

In some applications, it is necessary to fit data points with something other than a straight line.

$$y = X\beta + \epsilon$$



Residual vector

Any equation of this form is referred to as a **linear model**.

# Least-Squares Fitting of Other Curves

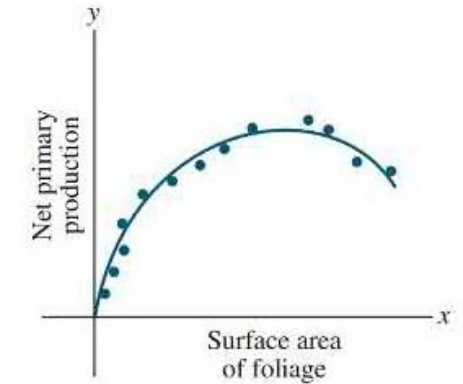
$$y = \beta_0 f_0(x) + \beta_1 f_1(x) + \cdots + \beta_k f_k(x)$$

## Example:

Suppose we wish to approximate the data by an equation of the form

$$y = \beta_0 + \beta_1 x + \beta_2 x^2$$

Describe the linear model that produces a "least-squares fit" of the data.



## Solution:

$$y_1 = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \epsilon_1$$

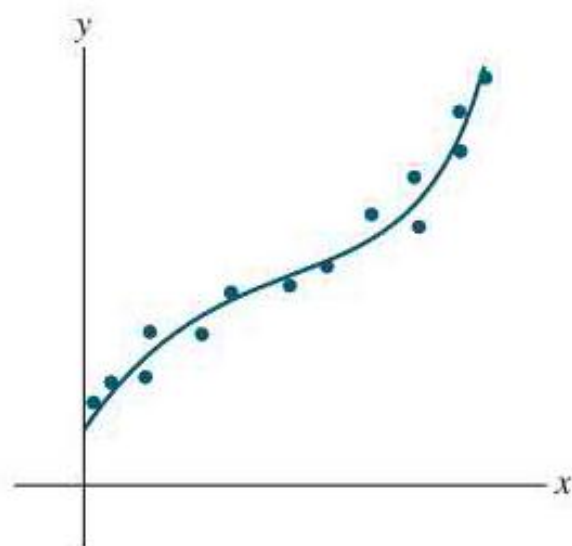
where  $\epsilon_1$  is the residual error between the observed value  $y_1$  and the predicted  $y$ -value  $\beta_0 + \beta_1 x_1 + \beta_2 x_1^2$ . Each data point determines a similar equation:

$$\begin{aligned} y_1 &= \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \epsilon_1 \\ y_2 &= \beta_0 + \beta_1 x_2 + \beta_2 x_2^2 + \epsilon_2 \\ &\vdots \\ y_n &= \beta_0 + \beta_1 x_n + \beta_2 x_n^2 + \epsilon_n \end{aligned}$$

It is a simple matter to write this system of equations in the form  $\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon}$ . To find  $X$ , inspect the first few rows of the system and look for the pattern.

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

$\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon}$



**FIGURE 6**

Data points along a cubic curve.

**EXAMPLE 4** If data points tend to follow a pattern such as in Figure 6, then an appropriate model might be an equation of the form

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3$$

Such data, for instance, could come from a company's total costs, as a function of the level of production. Describe the linear model that gives a least-squares fit of this type to data  $(x_1, y_1), \dots, (x_n, y_n)$ .

**SOLUTION** By an analysis similar to that in Example 2, we obtain

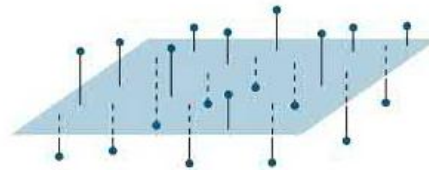
| Observation<br>vector                                                     | Design<br>matrix                                                                                                                                        | Parameter<br>vector                                                                            | Residual<br>vector                                                                                       |
|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix},$ | $X = \begin{bmatrix} 1 & x_1 & x_1^2 & x_1^3 \\ 1 & x_2 & x_2^2 & x_2^3 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 & x_n^3 \end{bmatrix},$ | $\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \beta_3 \end{bmatrix},$ | $\boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$ |



# Multiple Regression

$$y = \beta_0 f_0(u, v) + \beta_1 f_1(u, v) + \cdots + \beta_k f_k(u, v)$$

**EXAMPLE 5** In geography, local models of terrain are constructed from data  $(u_1, v_1, y_1), \dots, (u_n, v_n, y_n)$ , where  $u_j$ ,  $v_j$ , and  $y_j$  are latitude, longitude, and altitude, respectively. Describe the linear model based on (4) that gives a least-squares fit to such data. The solution is called the *least-squares plane*. See Figure 7.



**FIGURE 7** A least-squares plane.

**SOLUTION** We expect the data to satisfy the following equations:

$$\begin{aligned} y_1 &= \beta_0 + \beta_1 u_1 + \beta_2 v_1 + \epsilon_1 \\ y_2 &= \beta_0 + \beta_1 u_2 + \beta_2 v_2 + \epsilon_2 \\ &\vdots \\ y_n &= \beta_0 + \beta_1 u_n + \beta_2 v_n + \epsilon_n \end{aligned}$$

This system has the matrix form  $\mathbf{y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon}$ , where

$$\begin{array}{c} \text{Observation} \\ \text{vector} \end{array} \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \begin{array}{c} \text{Design} \\ \text{matrix} \end{array} \quad X = \begin{bmatrix} 1 & u_1 & v_1 \\ 1 & u_2 & v_2 \\ \vdots & \vdots & \vdots \\ 1 & u_n & v_n \end{bmatrix}, \quad \begin{array}{c} \text{Parameter} \\ \text{vector} \end{array} \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}, \quad \begin{array}{c} \text{Residual} \\ \text{vector} \end{array} \quad \boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

